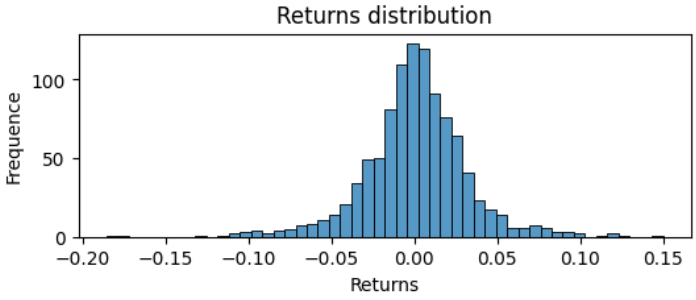
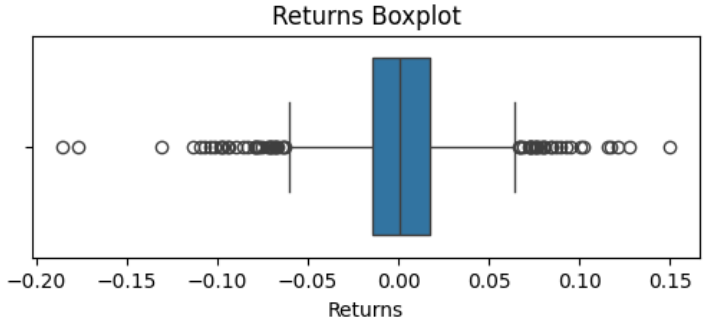
Selected stock: MercadoLibre (MELI)

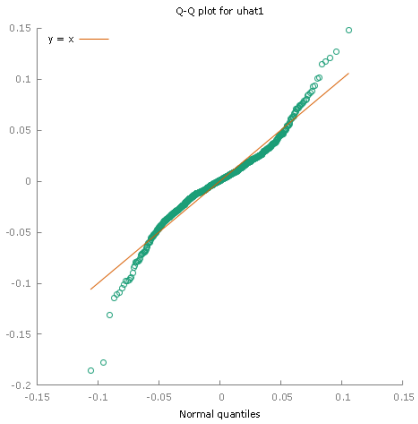
Time frame analysed: 2022-1-3 to 2026-1-15

Exercise 1



This distribution is not normal. Due to being mildly skewed to the left and heavy tails with a calculated excess Kurtosis of 3.9. Additionally, following graphs denote the significant outliers. Lastly, has a considerable Coefficient of Variation of 75%.





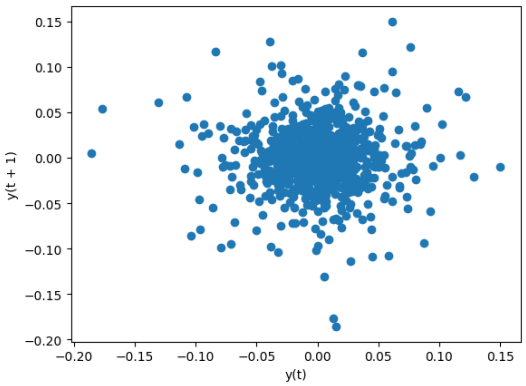
Mean	0.000425096
Median	0.000942007
Minimum	-0.184871
Maximum	0.149794
Standard deviation	0.0321890
C.V.	75.7218
Skewness	-0.297669
Ex. kurtosis	3.87367
5% percentile	-0.0506736
95% percentile	0.0500478
Interquartile range	0.0314935
Missing obs.	0

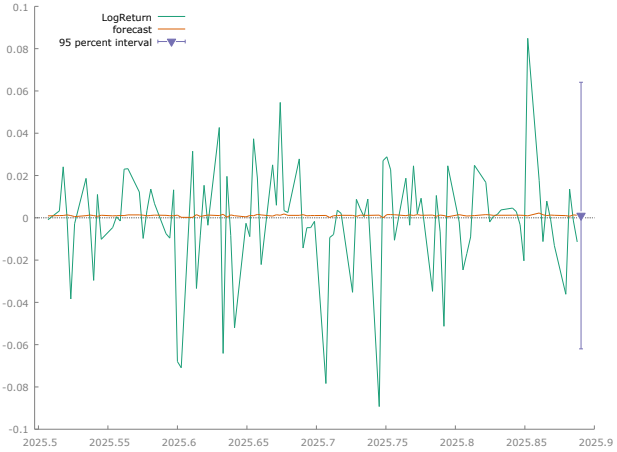
Considering this, in the parametric side we decided to test how a t-distribution fits.

But after reviewing results there is a big breach between the Value and Risk and Expected Shortfall of this t-distribution (-5,99% and -8,90%), and the ones evidenced by the non-parametric approaches (-17,60% and -18,11%). Consequently, it is needed to be modelled with non-parametric approaches.

Exercise 2

Reviewing autocorrelation of returns and an ARMA model capability to predict. By observing the scatterplot, we cannot denote any trend, it constitutes a cloud form that denies any autocorrelation between a return and its day before value.





For one step ahead prediction, the model calculates a value of 0,0010 but with a truly high standard error of 0,03216.

The high standard deviation indicates substantial variability, making the model's predictions unprecise. Furthermore, R squared and coefficients of the variables in the ARMA are very low. Even if the BIC value is fine, seems advisable just to use the mean in this case.

	coefficient	std. error	z	p-value
const	-0,00347318	0,0020332	-0,1708	0,8644
phi_1	0,0126276	0,744264	0,01697	0,9865
theta_1	0,00009999e-05	0,745982	0,000134	0,999
time	1,54365e-06	3,47167e-06	0,4446	0,6566

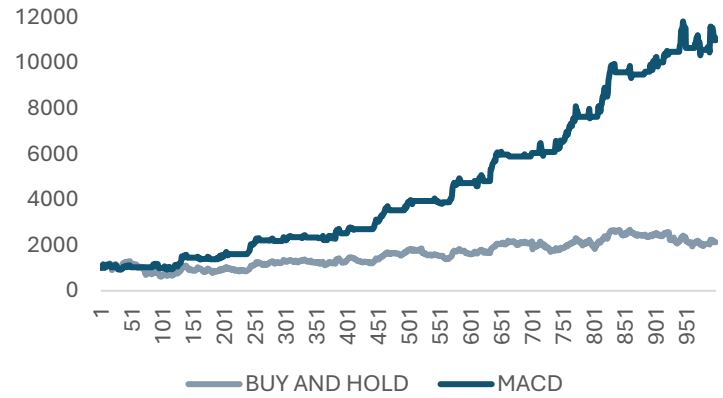
Mean dependent var	0,000425	**S.D. dependent var**	0,003219
mean of innovations	0,000011	**S.D. of innovations**	0,032167
R-squared	0,000374	**Adjusted R-squared**	-0,0016
Log-likelihood	2046,121	**Akaike criterion**	-4082,24
Schawrz criterion	-4057,634	**Hannan-Quinn**	-4072,89

Exercise 3

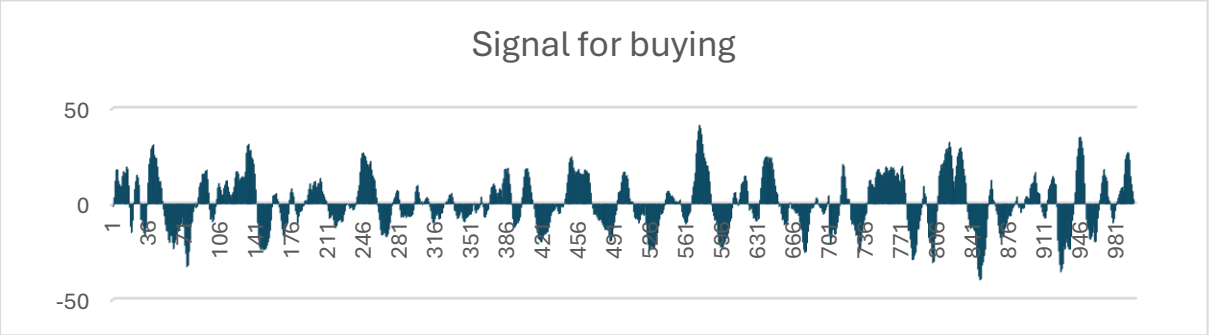
MACD-based trading strategy and its performance in MELI's case.

	BUY AND HOLD	MACD
MEAN	0,08%	**0,24%**
ST.DEV	50,28%	**34,14%**
SHARPE	0,001505	**0,007043**

MACD-based trading strategy outperforms in every analysed aspect the simpler buy and hold strategy.

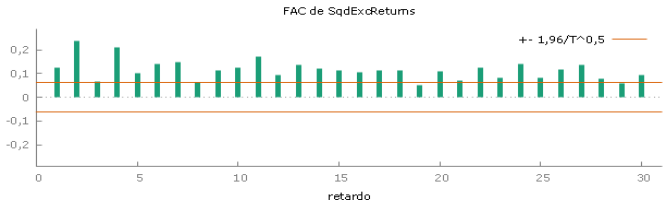


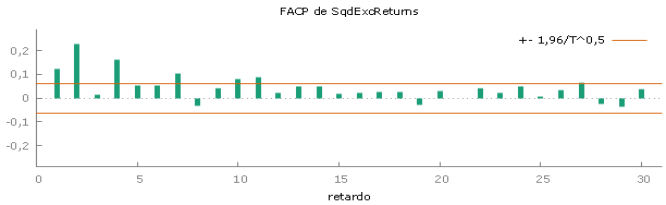
In this particular case, following MACD strategy is the best choice. As the movements on stock price allows such path.



Exercise 4

Squared excess returns seem autocorrelated with their past.

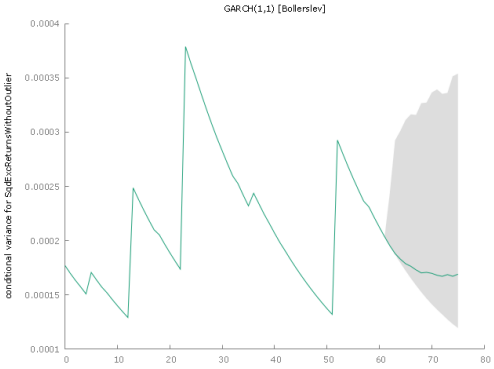




While analysing its dynamics, we first chose a GARCH with normal distribution. Our model gave both low p-values (<0,01) and Schwarz criterion (-9537,653), and a high chi-squared (306) which indicate that this model was appropriated at first instances. Conversely, when we compared the Unconditional Standard Error of 0,01532 and Standard deviation of the GARCH model of 0,00179772 we concluded that the model could not predict properly.

In this sense, we finally used GARCH t-variant and excluded outliers, which ended up giving us these results about the volatility which is reliable. In this stock and based in our analysis GARCH does seem to work properly.

	coefficient	std. error	z	p-value
const	0,0001827	1,87E-05	9,789	1,26e-022
omega	4,6584E-07	5,4801e-06	0,08499	0,9323
alpha	3,4336	36,74	0,09347	0,9255
beta	0,9563	0,0464	20,59	3,25e-094
LLIK	5806,279	AIC	-11602,6	
BIC	-11578,051	HQC	-11593,2	



Beta, the coefficient of past variance, is crucial to the prediction, with a low p-value and a convincing Z-score.